A Computational View of Medicine

Why AI will matter most where disease becomes a search problem

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Thesis. The most important medical use of advanced AI is not generic automation. It is high-context reasoning over difficult, patient-specific problems where better synthesis can change diagnosis, treatment choice, escalation, or trial matching. In these settings, disease is better understood as a computational burden distributed across time, modalities, and decision branches than as a static label alone.

Abstract

Artificial intelligence will not improve every part of medicine equally. Its highest-value role is in disease settings where the bottleneck is not data collection or workflow automation but high-context reasoning over multimodal evidence under uncertainty. We argue for a computational view of medicine that treats serious illness as a search problem over longitudinal records, pathology, imaging, genomics, patient-generated data, treatment history, and branching therapeutic options. The key question is not whether a model can produce clinically fluent text, but whether additional inference changes the next real decision for a specific patient. This constraint applies to clinicians and AI systems alike: both face the same search problem, and the quality of the search depends on the depth and structure of reasoning applied. We synthesize evidence from precision oncology, rare-disease genomics, antimicrobial discovery, sepsis surveillance, patient-reported outcomes, open-notes adoption, and N-of-1 medicine. Patient access to records is necessary but insufficient; because clinical reasoning is distributed across specialists and institutions—with the patient as the only persistent entity—data access becomes powerful only when patients and clinicians can repeatedly compute over the full timeline of illness. The result is a framework for directing AI toward cases where reasoning is the bottleneck, outcomes depend on synthesis quality and calibration, and patient-held data can become a force multiplier.

1 Introduction

In many of the cases that matter most in medicine, the limiting factor is not missing data. It is insufficient reasoning over data already available. The difficult work is synthesis: reconciling longitudinal evidence, deciding which signals matter, ranking plausible actions, and updating the model of a case as new information arrives.

This constraint is visible in everyday care. If a clinician had more time to think, would the next decision improve? For many routine cases, probably not. For high-burden cases—a confusing constellation of symptoms, an ambiguous genomic result, a cancer that has stopped responding—the answer is often yes. That makes reasoning a real clinical resource, and it makes reasoning scarcity a real clinical bottleneck.

Clinical reasoning can be carried out by clinicians, AI systems, or human-AI teams. The important question is not who reasons in the abstract, but whether additional reasoning changes the next meaningful decision for a patient. Under this view, advanced AI should be judged not by whether it produces fluent medical language, but by whether it adds useful reasoning capacity where the decision is difficult, high-stakes, and patient-specific. This extends Topol's case for human-machine medicine by placing computational leverage, rather than generic augmentation, at the center [34].

The constraint is not unique to machines. Physicians employ both rapid pattern recognition and slower deliberate analysis—dual-process reasoning—switching between modes depending on case complexity [7]. But time per patient is finite. Approximately 5% of US adults—over 12 million people—experience a diagnostic error in outpatient care each year [31], and an estimated 795,000 Americans suffer permanent disability or death annually from diagnostic errors, concentrated in cardiovascular events, infections, and cancer [24]. The "diagnostic timeout" in emergency medicine, a structured pause to reconsider the differential, reduces diagnostic error [7, 10]. Primary care consultation lengths range from 5 minutes to over 20 across healthcare systems [14]; if reasoning time is part of the bottleneck, that variation should predict quality.

Reasoning-focused AI systems can already participate in this deliberate layer of medicine. They can generate hypotheses, weight differentials, and integrate distributed evidence. In your own Frontiers study of DeepSeek R1, the model demonstrated structured clinical reasoning across diagnosis and management tasks, with error analysis showing that the main limitations were not absence of reasoning but specific failures such as anchoring, overthinking, and incomplete integration of conflicting data [22]. On complex diagnostic cases from the *New England Journal of Medicine*, GPT-4 included the correct diagnosis in 64% of cases, compared to 36% for medical-journal readers [19]. Med-PaLM 2 achieved 86.5% on USMLE-style questions and was preferred over physician answers on eight of nine clinical axes [32]. These benchmark results do not prove clinical utility on their own. They do show that added machine inference can improve the quality of clinical synthesis in some settings.

Reasoning as a clinical resource. The core question in advanced medical AI is not whether machines can reason in the abstract, but whether more reasoning—human, machine, or combined—changes the next real decision for a patient.

We call this a computational view of medicine. It treats disease management as the problem of converting fragmented, multimodal, longitudinal evidence into better patient-specific decisions under uncertainty. The argument of this paper is simple: disease classes differ sharply in how much more computation can improve outcomes, patient-held data changes where that computation can happen, and the highest-value uses of AI will concentrate in the cases where reasoning is the bottleneck.

Definition. A *computational view of medicine* treats disease management as the problem of converting fragmented, multimodal, longitudinal evidence into better patient-specific decisions under uncertainty.

2 A framework for computational leverage

We propose five properties that determine whether a disease class is likely to benefit from larger inference budgets:

1. **Search-space size**: how many plausible hypotheses, targets, interventions, or sequences must be evaluated.
2. **Data burden**: how many modalities and how much longitudinal context are required to reason well.
3. **Personalization**: how much the correct answer depends on the specific patient rather than the average protocol.
4. **Actionability**: whether better synthesis can materially alter diagnosis, escalation, treatment choice, or trial eligibility.
5. **Compute elasticity**: whether additional reasoning depth is likely to improve the result instead of merely restating the same answer more fluently.

These variables describe a single question: when does more reasoning change the next decision? Precision oncology, rare-disease genomics, resistant infection, critical care syndromes, and long-horizon chronic conditions rise to the top because they combine large search spaces, heterogeneous data, and decisions whose quality directly affects outcome. Diseases governed primarily by narrow protocols benefit more from reliable automation than from deeper inference.

Compute elasticity is the key discriminator. In low-elasticity domains, a clinician at normal pace already reaches the correct decision and more thought adds little. In high-elasticity domains, additional reasoning can shift the differential, reorder treatment options, improve calibration, or surface a branch that would

Domain	Search	Data	Pers.	Action	Elast.	Interpretation
Precision oncology	5	5	5	5	5	Dense multimodal evidence and many branching intervention paths make more inference directly decision-relevant.
Rare-disease genomics	5	5	5	4	5	Large hypothesis spaces and sparse distributed clues create unusually high value for long-context reasoning.
Resistant infection	4	4	4	5	4	The answer depends on organism, host state, prior exposure, and time-sensitive escalation.
Sepsis / ICU	4	5	4	5	4	Continuous streaming data create a high-burden monitoring problem in which timing dominates outcome.
Chronic inflammatory disease	3	4	4	4	3	Longitudinal symptom and biomarker integration can improve timing and sequencing, but action spaces are narrower.
Routine protocol care	1	2	1	2	1	Most value lies in reliable execution and logistics rather than larger inference budgets.

Table 1: Illustrative scoring matrix for computational leverage. Scores are conceptual rather than empirical, but they make the framework explicit and falsifiable.

otherwise be missed. AI does not replace clinical reasoning in these cases. It extends the reasoning budget available before action is taken.

This is where the abstract framework meets clinical work. A molecular tumor board may have minutes to search a decision space of drugs, trials, biomarkers, and prior-line contingencies. A geneticist may have one review session to integrate years of notes, family history, imaging, and genome data. A primary care physician managing multimorbidity must often act before the full interaction space has been searched. We refer to this gap between what could in principle be searched and what is actually searched in routine care as the *compute deficit*. The largest returns from medical AI will come from shrinking that gap in domains where better search changes patient care.

Table 1 should therefore be read as a proposed allocation heuristic, not as a finished empirical scorecard. It makes the framework explicit enough to test. *Search-space size* can be approximated by counting plausible differentials or candidate interventions; *data burden* by the number of modalities and timescales required for a defensible decision; and *compute elasticity* by asking whether more reasoning time improves ranking quality, calibration, or branch selection. Figure 1 addresses a different part of the argument: not leverage itself, but urgency. The concentration of serious diagnostic harm in the same broad domains suggests that the reasoning gap is a measurable public-health problem rather than a theoretical one.

Why prevalence is not enough. Common diseases are not automatically the best targets for large inference budgets. The relevant question is whether more synthesis can change a meaningful patient-specific decision. That is why prevalence rankings and compute-opportunity rankings diverge.

3 Why patient-held data matter

Clinical reasoning in modern healthcare is distributed across general practitioners, specialists, pathologists, radiologists, and multidisciplinary teams. Each referral adds expertise, but each handoff also risks losing state.

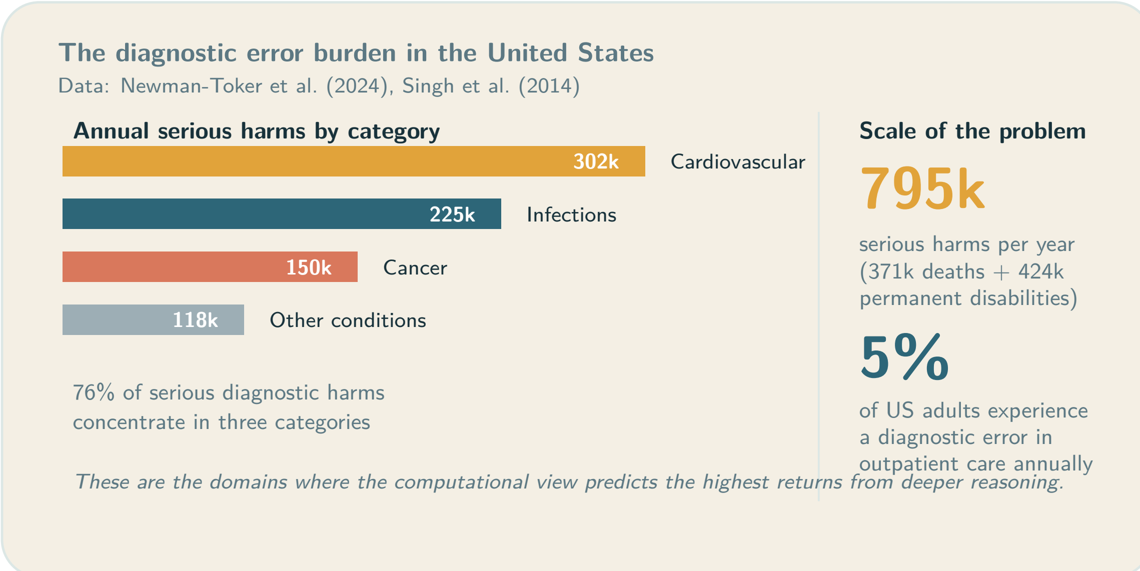

Figure 1: The diagnostic error burden concentrates in exactly the domains where the computational view predicts the highest compute elasticity. Data from Newman-Toker et al. [24] and Singh et al. [31].

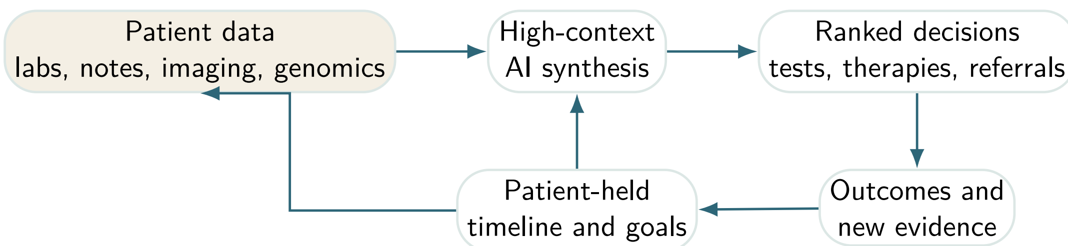

Figure 2: The patient-compute loop. In high-burden disease, value comes from repeated synthesis over the full timeline rather than one-shot question answering.

The patient is the only entity that persists across all of those encounters. That fact matters computationally. If the case is going to remain coherent across time and institutions, the longitudinal state of the problem has to be reconstructable around the patient.

This is why patient-held data matters. It is not only a matter of access rights or transparency. It changes where medical reasoning can live. Once records can be assembled across encounters, institutions, and modalities, computation can follow the patient rather than remaining trapped inside any one clinic, hospital, or specialty workflow.

OpenNotes showed that patients value direct access to clinicians' notes and that transparency can improve engagement [8]. Patients also shared those notes with family members and caregivers, turning the record into a coordination tool rather than a private institutional artifact [15]. Patient-generated data—symptom tracking, wearables, and home monitoring—extend that record beyond formal encounters [25]. The longitudinal case history is therefore no longer only what the institution stores. It is increasingly what can be assembled around the patient over time.

Access alone, however, is not enough. Precision medicine remains incomplete without patients as participants in how data are interpreted and used [35]. AI for shared decision-making must improve deliberation, not merely generate recommendations [12]. Integration of patient-generated health data is not only a technical problem but a clinical and organizational one [16]. Early work suggests that patients can already use large language models on open notes to interpret their records [29]. Your DeepSeek R1 paper adds a second piece to that picture: if models are capable of genuine clinical reasoning over heterogeneous medical information, then the computational layer does not need to sit only with the clinician or only with the institution [22]. It can be shared. Patients who have access to their data, sufficient context, and enough health literacy to work with these tools can use them to prepare, interrogate, and structure their case before seeing a clinician, while the clinician can use the same computational layer to refine or challenge that reasoning. The larger implication is architectural: without patient-held continuity of state, longitudinal computation cannot reliably follow the case; with it, repeated reasoning over the full illness trajectory can become a shared capability between patients and clinicians.

4 Evidence from the highest-leverage domains

The domains below are not examples at random. They are cases in which the framework above predicts high computational leverage and where current evidence already supports that prediction to varying degrees. The logic is the same in each case: the clinical bottleneck is a search problem over distributed evidence, more reasoning can change the next action, and the value of better synthesis is large enough to matter clinically.

4.1 Precision oncology

Cancer is already a computational problem. Molecular diagnosis, resistance interpretation, treatment sequencing, and trial matching require synthesis across pathology, imaging, genomics, prior therapy, and time. Personalized cancer vaccines make this explicit: identifying patient-specific tumor mutations, ranking neoantigens, and manufacturing a bespoke intervention. Sahin and colleagues showed that individualized RNA mutanome vaccines mobilize poly-specific immunity against melanoma [28]. AlphaFold expands the search surface further by making protein consequence analysis tractable across large molecular spaces [18].

Compute sits inside the decision itself: which mutation matters, which escape mechanisms are plausible, which combination is worth pursuing, which trial is defensible for this patient now. The burden of reasoning has become part of the disease burden.

A single patient may generate surgical pathology, radiology progression patterns, panel sequencing, circulating tumor DNA, treatment toxicities, and eligibility-relevant comorbidities. The computational task is to rank drivers, resistance hypotheses, and therapeutic branches while making uncertainty visible. The leverage comes not from generating prose about cancer but from compressing a large search problem into a smaller set of defensible next moves.

4.2 Rare disease genomics

The problem is not only finding a variant but aligning phenotype, pedigree, prior negative workup, and the literature fast enough to affect care. Rare disease is one of the clearest examples of why patient-held continuity matters: the relevant clues are often accumulated across years, institutions, and specialties. Wojcik and colleagues showed in the *New England Journal of Medicine* that genome sequencing materially improved diagnosis for suspected rare disease [36]. This is a canonical compute-heavy domain: large hypothesis spaces, sparse signal, fragmented records, and a direct payoff when reasoning succeeds. The cost of failure is now quantified: rare-disease patients accumulate average healthcare costs of €27,000 over their diagnostic odyssey, compared to €3,600 for matched controls, and avoidable costs of delayed diagnosis in the US range from $86,000 to over $517,000 per patient [9].

4.3 Resistant infection and antibiotic discovery

Antimicrobial resistance presents a two-layer search problem. The first is patient-specific: organism identity, susceptibility, host state, prior exposure, organ function, and drug interactions. The second is discovery: searching chemical space for compounds with novel activity. In the clinic, better reasoning changes which therapy is started, escalated, or abandoned. In the lab, it changes which candidates are worth testing at all. Stokes and colleagues used deep learning to identify halicin, accelerating antibiotic discovery in regions of chemical space invisible to standard pipelines [33]. More compute changes action in both settings: regimen ranking in the clinic and candidate exploration in the lab.

4.4 Sepsis and critical care

In critical care the challenge is time. Dozens of measurements update continuously, and the relevant signal is distributed across trends rather than single observations. This makes sepsis and deterioration surveillance a different kind of search problem from oncology or genetics: less about finding the right branch in a large static space, more about detecting when the patient has moved onto a dangerous branch quickly enough to intervene. A machine-learning sepsis early-warning system deployed across multiple hospitals was associated with faster antibiotic administration and lower mortality [1]. Deep learning on electronic health records can scale across broad clinical prediction tasks using the raw longitudinal record [27], and pragmatic deterioration surveillance can attach to live escalation pathways [23]. These results establish that longitudinal records contain actionable signal and that systems which hold more context together can improve timing-sensitive decisions.

4.5 Psychiatry and mental health

Psychiatry is computational because the organ of interest is computationally complex, but also because the clinical categories themselves are messy. Comorbidity across psychiatric diagnoses is high, symptom boundaries are unstable, and many patients do not fit neatly inside a single disorder label. In practice, a clinician deciding what to do next in depression, bipolar disorder, ADHD, or anxiety is often not solving a clean classification problem. The task is to understand a particular person's pattern of suffering, functioning, prior treatment response, context, and trajectory well enough to choose the next useful move.

That makes psychiatric care, in a concrete sense, a sequential decision problem. Each intervention is an action taken under uncertainty: start a medication, stop one, increase a dose, change a therapy target, alter environmental structure, test a behavioral strategy, or wait for more signal. The next state of the patient depends partly on that action and partly on dynamics that are difficult to observe directly. When care works well, whether in psychotherapy, pharmacotherapy, or a combination, it is because the clinician and patient find a better next action for the current state.

This section is more inferential than oncology, rare disease, or sepsis, but it extends the same framework rather than departing from it. First-line antidepressants fail in a large share of patients, and standard care often proceeds by serial trial and error over months. Pharmacogenomics offers one example of computation reducing that search burden: cytochrome P450 variation can identify medications that are more likely to be poorly metabolized before they are prescribed [13]. The point is not that genotype solves psychiatry. It is that psychiatric treatment already contains a branching search problem in which better synthesis can eliminate bad branches earlier.

The same structure appears in psychotherapy. Cognitive behavioral therapy is, in practice, an effort to map the loops that keep a disorder stable: triggers, beliefs, behaviors, and reinforcing consequences [5].

That is close to an action-selection problem over state. A therapist is trying to identify which intervention is most likely to move the patient from a maladaptive local equilibrium toward a better one, given the current pattern of thought, behavior, and context. Tolstoy's line that "every unhappy family is unhappy in its own way" captures something clinically relevant here: psychiatric illness is often idiosyncratic in its exact configuration, even when it resembles a familiar syndrome. Under a computational view, psychiatry belongs in the portfolio of high-burden reasoning problems precisely because more context, better within-person modeling, and better action selection can change the next therapeutic move.

4.6 Patient-reported outcomes and symptom monitoring

Patient-reported outcomes provide a useful bridge case because they show that high-value computation is not limited to genomics, molecular medicine, or the ICU. Symptoms, side effects, and tolerability often live outside the hospital until severe enough to trigger contact. The computational task is therefore not simply to collect more symptom reports, but to infer from noisy, time-distributed signals whether a patient is deteriorating, failing therapy, or developing treatment intolerance before the situation becomes acute. Electronic patient-reported symptom monitoring during cancer treatment improved overall survival, likely because it made those weak longitudinal patterns actionable earlier [3, 4]. In the computational paradigm, symptom streams become first-class inputs into the case model, and their value comes from repeated integration over time. The broader point is that as more of the clinically relevant state lives between visits, systems that can repeatedly synthesize those signals can change the timing and quality of intervention.

5 N-of-1 medicine and personal computation

N-of-1 medicine is the natural clinical endpoint of the computational view. If the central problem is patient-specific reasoning over time, then the scientifically relevant unit is not only the cohort average but the evolving individual case. Schork argued that precision medicine must be understood at the level of the individual [30]. Guyatt laid the foundation for randomized trials in individual patients [11], and Lillie argued that N-of-1 trials align with the actual endpoint of clinical practice: optimizing care for one person [20]. Augustine proposed a roadmap for N-of-1 therapies covering scientific, operational, and regulatory requirements [2]; Jonker outlined how N-of-1 medicine could reshape treatment evaluation for rare and heterogeneous disease [17]. Taken together, these contributions move personalization from slogan to scientific unit.

Longitudinal self-measurement provides parallel evidence. Integrative personal omics profiling reveals dynamic molecular phenotypes within a single individual over time [6]. Dense personal data clouds can monitor health transitions with temporal resolution far exceeding episodic care [26]. Biology, symptoms, and risk are often more legible in longitudinal within-person data than in sparse snapshots. The clinical payoff is not merely richer description. It is better hypothesis testing, better adaptation of therapy, and better discrimination between fluctuation and meaningful state change within a person.

Public self-experimentation—highly instrumented programmes such as Bryan Johnson's—is not clinical evidence. It is better understood as a demand signal: people want dense longitudinal understanding of their own bodies and health trajectories. The scientific path forward is not informal self-optimization, but audited N-of-1 workflows that combine patient goals, validated measurements, explicit hypotheses, and clinically accountable reasoning.

6 Context engineering within health AI

One practical implication follows from the computational view: many reasoning failures in medicine are context failures. Clinicians often reason with incomplete longitudinal state because the relevant record is fragmented across institutions, specialties, and time. Patients often reason with incomplete mechanistic context because they do not have access to the clinician's full model of how a disorder works, what matters, and which branches are under consideration. AI systems fail for the same reason when they are asked to reason over a thin snapshot instead of the case as it actually unfolds. In all three settings, poor reasoning is frequently the downstream effect of poor context.

Context engineering, in this setting, means making the patient case legible enough that clinicians, patients, and AI systems can reason over the same evolving state. That requires more than document retrieval. The relevant context has to preserve chronology, distinguish observed evidence from inferred conclusions, carry forward prior treatments and their effects, represent uncertainty explicitly, and include the patient goals and constraints that determine which next action is actually good. A useful health context is not a pile of records. It is a structured case model that keeps the right facts, timelines, hypotheses, and decision branches available at the moment of reasoning.

This matters for both patients and clinicians. If patients are given access to the relevant data sources in usable form, they can run computation over their own cases before and between visits, rather than entering the clinic with only memory and fragments. If clinicians receive that context back in structured form, they do not need to repeatedly reconstruct the case from scratch. The point is not to replace either side with an AI system, but to let the computational layer be shared: the patient can contribute lived state, symptoms, goals, and continuity; the clinician can contribute mechanistic knowledge, risk judgment, and accountability; and AI tools can help synthesize the combined context into better next-step reasoning.

Tables 2 and 3 and Figure 3 make this concrete. Across disease areas, the same question recurs: what context has to be present for the next decision to improve? The answer differs by domain, but the design principle is stable. Health problems become more computational when the right longitudinal context is assembled, structured, and repeatedly available to all of the agents reasoning about the case.

7 Evaluation and endpoints

The computational view should change how health AI is evaluated. Most current evaluation still privileges prediction quality in isolation: accuracy on benchmark questions, area under the curve on retrospective labels, or ranking quality on a fixed dataset. Those measures are useful, but they are not the right target for this paper's thesis. The claim here is not simply that models can predict. It is that better computation can improve the quality of reasoning at the point where care can still change direction.

That requires distinguishing three levels of performance. *Prediction quality* asks whether the model's output matches a label or outcome. *Synthesis quality* asks whether the system assembled the relevant evidence, represented uncertainty, and reasoned coherently over competing possibilities. *Decision quality* asks whether the resulting action—test, referral, escalation, treatment change, or watchful waiting—was better because more reasoning occurred. A system can improve one level without improving the next. The computational thesis is strongest when synthesis quality and decision quality improve together.

Calibration is therefore as important as discrimination. In high-stakes medicine, a system that says "60% diagnosis A, 25% diagnosis B, and here is the test that best separates them" may be more clinically useful than one that outputs only a top-ranked answer, even at identical accuracy. More reasoning will sometimes leave the leading diagnosis unchanged while improving uncertainty estimation, clarifying missing evidence,

Domain	Core data	How computation is leveraged	Representative evidence
Precision oncology	Genomics, pathology, imaging, treatment history	Variant interpretation, resistance modeling, neoantigen ranking, regimen sequencing, trial matching	Sahin et al. [28]; Jumper et al. [18]
Rare disease	Phenotype timeline, pedigree, genome sequencing, prior workup	Cross-linking phenotype ontologies, literature, family history, and candidate variants	Wojcik et al. [36]
Resistant infection	Microbiology, susceptibilities, organ function, prior antibiotic exposure	Better ranking of therapies and faster search across compound or regimen space	Stokes et al. [33]
Sepsis / ICU	Streaming vitals, labs, notes, medication events	Earlier escalation from weak, time-distributed patterns across the longitudinal record	Adams et al. [1]; Rajkomar et al. [27]; Moses et al. [23]
Patient-reported symptoms	Patient-reported outcomes, treatments, toxicities, goals	Detecting deterioration and treatment intolerance before they become acute	Basch et al. [3]; Basch et al. [4]
Patient-held records	Notes, meds, labs, imaging reports, symptoms, goals	Converts raw data access into usable reasoning leverage before and between encounters	Delbanco et al. [8]; Salmi et al. [29]

Table 2: Context requirements for high-value disease areas under a computational lens.

or ruling out unsafe branches. Those gains matter because they influence whether a clinician orders a confirmatory test, refers urgently, escalates therapy, or waits.

Evaluation should also move from static outputs to workflow effects. The relevant question is not only whether an answer was correct, but whether the system changed the next meaningful step in a defensible way. In a rare-disease workup, did it shorten the path to diagnosis or reduce unnecessary procedures? In oncology, did it improve treatment ranking or trial matching? In sepsis, did it improve timing without producing alert burden that overwhelms the team? In patient-facing systems, did it improve comprehension, question quality, continuity across visits, or the ability to arrive with a usable case model rather than scattered documents?

This implies a different research programme. Prospective studies should test computational systems at real decision points rather than only on retrospective corpora. They should measure not only correctness but branch quality: whether the system improved calibration, reduced time spent on the wrong branch of care, made uncertainty more usable, and strengthened patient-clinician preparation before action. A computational evaluation framework is therefore not opposed to conventional model metrics. It places them inside a stricter question: did added reasoning improve the next decision for this patient in this setting?

Future patient personas

Patient	Data substrate	Computational leverage
Asha, 58 metastatic NSCLC	Tumour DNA/RNA, pathology, CT trends, prior lines, toxicities, performance status	Variant prioritization, resistance hypotheses, neoantigen ranking, and live trial matching change the next therapeutic branch.
Mateo, 6 undiagnosed neuro-developmental syndrome	Phenotype timeline, EEG, MRI, family history, genome sequencing, prior negative workup	Phenotype-variant ranking and literature synthesis narrow candidate mechanisms and guide targeted confirmatory testing.
Leila, 34 Crohn's disease on biologics	Symptoms, calprotectin, medication adherence, wearables, sleep, diet, clinician notes	Within-person pattern detection turns fragmented self-tracking into actionable flare prediction and better-timed escalation.

Figure 3: Concrete personas for a computational health stack. The same design principle appears across specialties: assemble the full timeline, structure the context, compute over it repeatedly, and use the result to improve the next decision.

8 Risks and constraints

The computational view sharpens the need for evidence and oversight because it directs AI toward the parts of medicine where mistakes matter most. High-burden disease is exactly where hallucinated synthesis, hidden bias, brittle prompting, and false confidence are most dangerous. The core safety problem is not generic model error. It is wrong computation at the wrong branch point: a missed escalation in sepsis, a plausible but incorrect molecular rationale in oncology, a false narrowing of the differential in rare disease, or a confident but poorly grounded recommendation that closes off better options. Reasoning systems in these settings must therefore be auditable, source-grounded, and explicit about what they do not know.

Context failure is itself a safety problem. If the paper's central claim is that better reasoning requires better context, then incomplete or malformed context can create the appearance of intelligence while degrading the actual decision. A model given an unstructured fragment of the chart, a patient missing key prior results, or a clinician working without the full longitudinal state may each arrive at an internally coherent but clinically wrong answer. The danger is not only absence of information but corrupted state: chronology lost, evidence confused with inference, prior treatment failures omitted, or patient goals excluded from the reasoning process.

There is also a risk of overextending the computational frame. Not every disease is compute-elastic. Some problems remain dominated by biology, logistics, staffing constraints, or protocol execution rather than reasoning depth. In those settings, adding more inference may create noise, delay, or false sophistication rather than benefit. The point of the framework is precisely to avoid spending the largest inference budgets where more reasoning cannot change the next meaningful decision.

Because this paper argues for a shared computational layer across patients, clinicians, and AI systems, trust calibration matters on all sides. Clinicians may over-trust fluent synthesis and fail to inspect the evidence

Study type	What changed	Why it matters for the thesis
Pragmatic randomized clinical trial in cardiovascular screening	AI-enabled ECG alerts were tested as an intervention rather than a retrospective classifier [21]	Supports the claim that computational systems can change downstream decisions and outcomes, not merely generate scores.
Pragmatic cluster-randomized deterioration trial	Real-time surveillance for patient deterioration was tested across live inpatient settings [23]	Strengthens the case that high-context clinical surveillance can function as an operational care system.
Patient-record sharing study	Patients with transparent notes shared them with family and caregivers, extending the practical reach of the record [15]	Shows that once patients have access to records, those records begin acting as coordination infrastructure.
Patient-side LLM proof of concept	Patients used large language models on open notes to better understand and work with their records [29]	Gives direct support to the argument that personal compute can become part of the patient toolkit.

Table 3: Evidence that context-rich computational systems can extend beyond retrospective prediction.

base. Patients may over-trust tools that help them reason over records without understanding the limits of those tools. Institutions may use apparent reasoning performance as a substitute for accountability. Good systems should therefore surface provenance, preserve uncertainty, and make it easy to inspect how a recommendation was constructed rather than merely presenting a persuasive answer.

Finally, if compute meaningfully improves outcomes, access to compute becomes part of health equity. A health stack that gives patients records without usable synthesis leaves too much leverage with the institution; a health stack that offers synthesis only to already well-resourced patients may widen disparities further. The practical constraint is not only whether better computation is possible, but whether it is broadly, safely, and accountably available.

9 Conclusion

Medicine is already a computational process. Evidence-based medicine formalized that process at the population level. The computational view proposed here formalizes it at the patient level and argues that a large share of avoidable harm in complex disease comes from the gap between what could be searched, synthesized, and updated for a patient and what is actually computed in routine care.

The diseases that matter most under this view are not simply the most prevalent. They are the ones that become search problems at patient resolution: precision oncology, rare-disease genomics, resistant infection, selected critical care settings, and chronic relapsing disease. In these domains, better reasoning can change the next branch in care.

The deeper implication is structural. If reasoning has historically been scarce and expensive—bound to the time, memory, and attention of individual clinicians—then cheap, portable computation changes not only the quality of individual decisions but the architecture of medicine itself. A computational view of medicine therefore does two things at once: it shows where additional inference can genuinely improve outcomes, and it points toward a health system in which understanding becomes a shared, longitudinal, patient-held capability rather than a privilege of the institution.

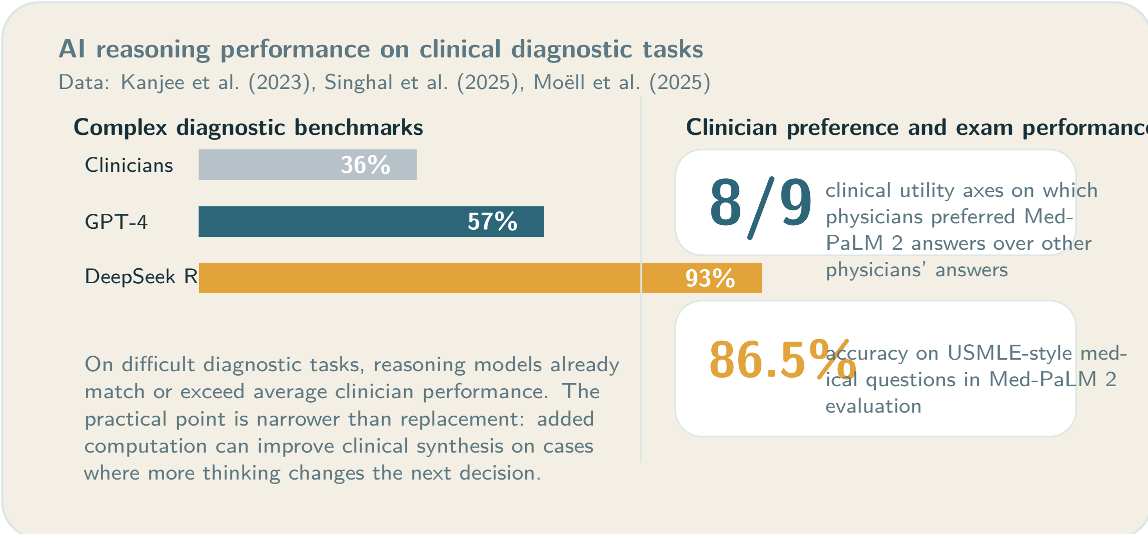

Figure 4: Reasoning-focused AI systems already perform strongly on difficult diagnostic and medical benchmarks. Data from Kanjee et al. [19], Singhal et al. [32], and Moëll et al. [22]. The clinical relevance is practical: in some cases, more computation improves the quality of synthesis.

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Domain	Primary endpoint candidates	Why these endpoints fit the computational thesis
Rare disease	Time to diagnosis; diagnostic yield; avoided procedures	Measures whether synthesis reduced time spent on the wrong branch of care.
Precision oncology	Time to molecular recommendation; trial-match yield; change in treatment ranking; time to next line	Measures whether computation improved branching decisions rather than documentation alone.
Sepsis / ICU	Time to escalation; antibiotics or intervention timing; false alerts per patient-day; mortality or ICU transfer	Measures whether longitudinal synthesis improved timing without overwhelming workflow.
Chronic disease / N-of-1 care	Time to flare detection; within-person treatment response; symptom burden; adherence to preferred regimen	Measures whether repeated computation improves individualized management over time.
Patient-facing record synthesis	Patient comprehension; visit preparedness; question quality; fewer duplicated tests; continuity across sites	Measures whether access plus reasoning improved the practical utility of records.

Table 4: Evaluation endpoints aligned with a computational view of medicine. The goal is to measure whether reasoning improves synthesis and decisions, not only whether predictions look accurate offline.

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